

Expansion Tank System: Process, Flow & Optimization Overview

Purpose:

Expansion tanks absorb thermal expansion in closed-loop hydronic systems. They prevent overpressure by compensating for volume changes when water heats up.

System Start-to-Finish Flow:

1. ****System Startup & Pressurization****
 - Closed-loop systems are filled with water and pressurized to static pressure (typically 12–25 psi).
 - Expansion tank is pre-charged with air (matching system pressure when cold).
2. ****Thermal Expansion During Heating****
 - As water heats and expands, pressure increases.
 - Extra volume is pushed into the expansion tank, compressing internal air bladder.
3. ****Pressure Control****
 - Bladder compresses to absorb volume, preventing pressure spikes.
 - Relief valves remain closed if expansion tank is sized and charged properly.
4. ****Cooling & Volume Reversal****
 - As system cools, water contracts and re-enters the system from the tank.
 - Air bladder re-expands to normal charge.

Key Components & Definitions:

- ****Bladder Tank****: Has internal diaphragm to separate air and water.
- ****Pre-Charge Valve****: Allows air pressure setting before fill.
- ****System Pressure Gauge****: Verifies proper tank pressure.
- ****Isolation Valve****: Used to isolate tank during service.
- ****Relief Valve****: Safety measure in case of pressure overage.

Example Flow:

> In a 3-story commercial building with a 50-psi hot water loop, a 40-gallon bladder tank is pre-charged to 22 psi. As the loop heats up from 70°F to 180°F, pressure increases and excess water enters the tank, stabilizing pressure without triggering the relief valve.